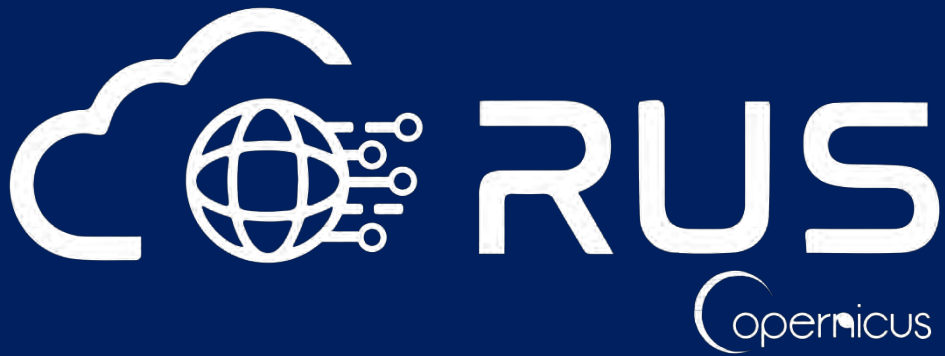
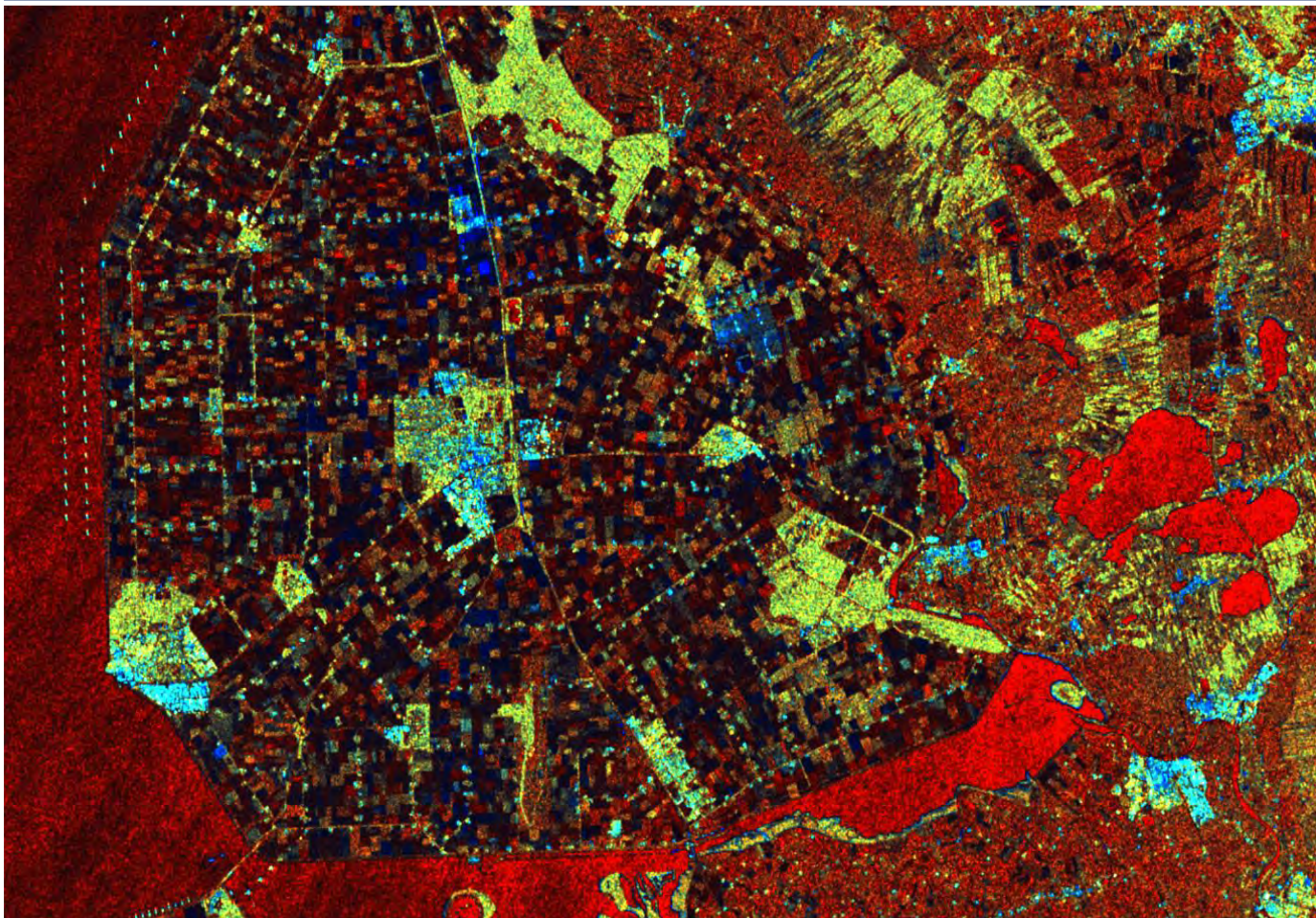




RUS

Copernicus



TRAINING KIT – PY01

SENTINEL-1 PROCESSING USING SNAPPY



Research and User Support for Sentinel Core Products

The RUS Service is funded by the European Commission, managed by the European Space Agency and operated by CSSI and its partners.

Authors would be glad to receive your feedback or suggestions and to know how this material was used. Please, contact us on training@rus-copernicus.eu

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1 Introduction to RUS

The Research and User Support for Sentinel core products (RUS) service provides a free and open scalable platform in a powerful computing environment, hosting a suite of open source toolboxes pre-installed on virtual machines, to handle and process data derived from the Copernicus Sentinel satellites constellation.

In this tutorial, we will employ RUS to learn how to process Copernicus data (Sentinel-1) using the Python *snappy* module.

2 Sentinel-1 – background



A few years ago, the European Union (EU) started an ambitious program, Copernicus, which includes the launch of a new family of earth observation satellites known as Sentinels. The Sentinel-1 mission is the European Radar Observatory for the Copernicus joint initiative of the European Commission (EC) and the European Space Agency (ESA). The Sentinel-1 mission

comprises a constellation of two polar-orbiting satellites, operating day and night and performing C-band Synthetic Aperture RADAR (SAR) imaging in four exclusive modes with different resolution (down to 5 m) and coverage (up to 400 km). It provides dual polarisation capability, very short revisit times and rapid product delivery. For each observation, precise measurements of spacecraft position and attitude are available.

In this exercise you will learn how to use *snappy* to apply a standard pre-processing chain and produce Sentinel-1 analysis-ready data. The analysis will be implemented using Python code that can be found in this Jupyter Notebook. Although highly recommended, the exercise DOES NOT require any Python programming skills and can be followed by any participant. You will not be asked to write any code but to follow the methodology and understand the processing steps.

3 Training

Approximate duration of this training session is **one** hour.

The Training Code for this tutorial is PY01. If you wish to practice the exercise described below within the RUS Virtual Environment, register on the RUS portal and open a User Service request from Your RUS service > Your dashboard.

3.1 Data used

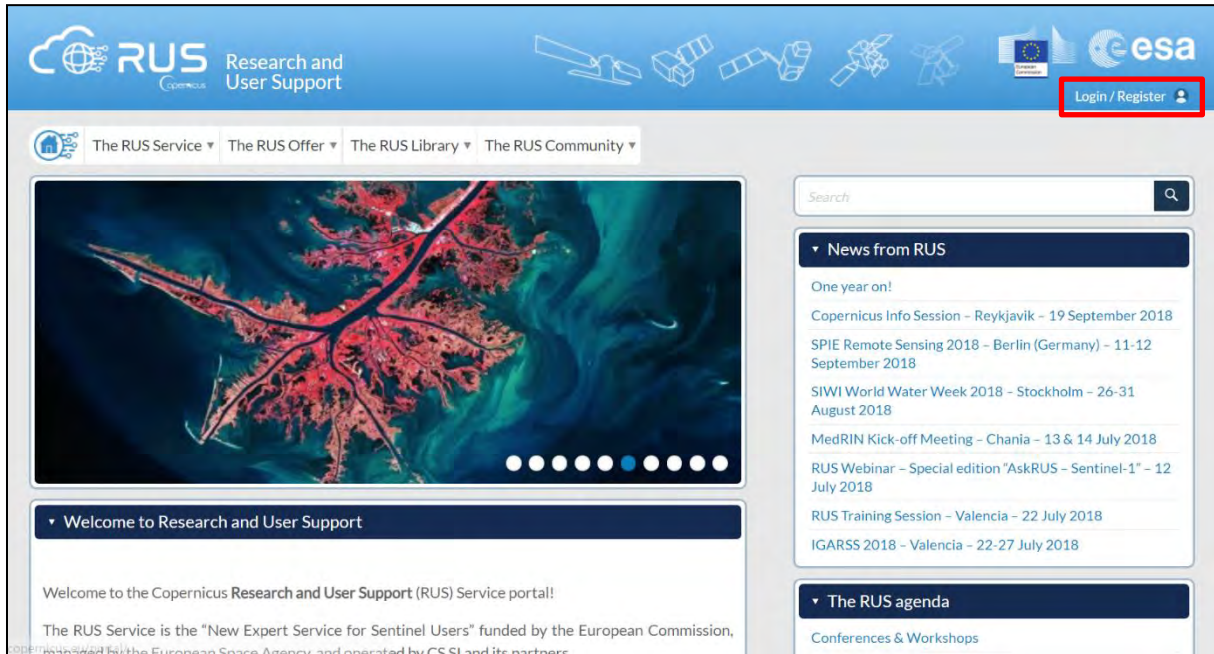
- 1 Sentinel-1 image acquired during April 2020
- Pre-processed data stored locally
`@/shared/Training/PY01_Sentinel1Processing_snappy/AuxData/`

3.2 Software in RUS environment

Internet browser, JupyterLab, Python, Anaconda, *snappy*

4 Register to RUS Copernicus

To repeat the exercise using a RUS Copernicus Virtual Machine (VM), you will first have to register as a RUS user. For that, go to the RUS Copernicus website (www.rus-copernicus.eu) and click on **Login/Register** in the upper right corner.



Select the option **Create my Copernicus SSO account** and then fill in ALL the fields on the **Copernicus Users' Single Sign On Registration**. Click **Register**.

Within a few minutes you will receive an e-mail with activation link. Follow the instructions in the e-mail to activate your account.

You can now return to <https://rus-copernicus.eu/>, click on **Login/Register**, choose **Login** and enter your chosen credentials.

Login / Register

The registration system to access the RUS service platform has moved toward the COPERNICUS Single Sign On authentication server.

- New Users who have not yet registered to the RUS portal shall first create a COPERNICUS SSO account.

Note that your Copernicus SSO account will be activated only after the reception of the third email sent by the Copernicus service. We advise you to consult [this document](#) and [this page](#) to facilitate your registration procedure.

REGISTER COPERNICUS SSO account

Users who already have a COPERNICUS SSO account can login here:

Login

Close

Credentials

CDS-SSO ID

Password

Max Idle Time

half a day

Max Session Time

Until browser close

Login

Reset

[Forgot your password?](#)

Upon your first login you will need to enter some details. You must fill all the fields.

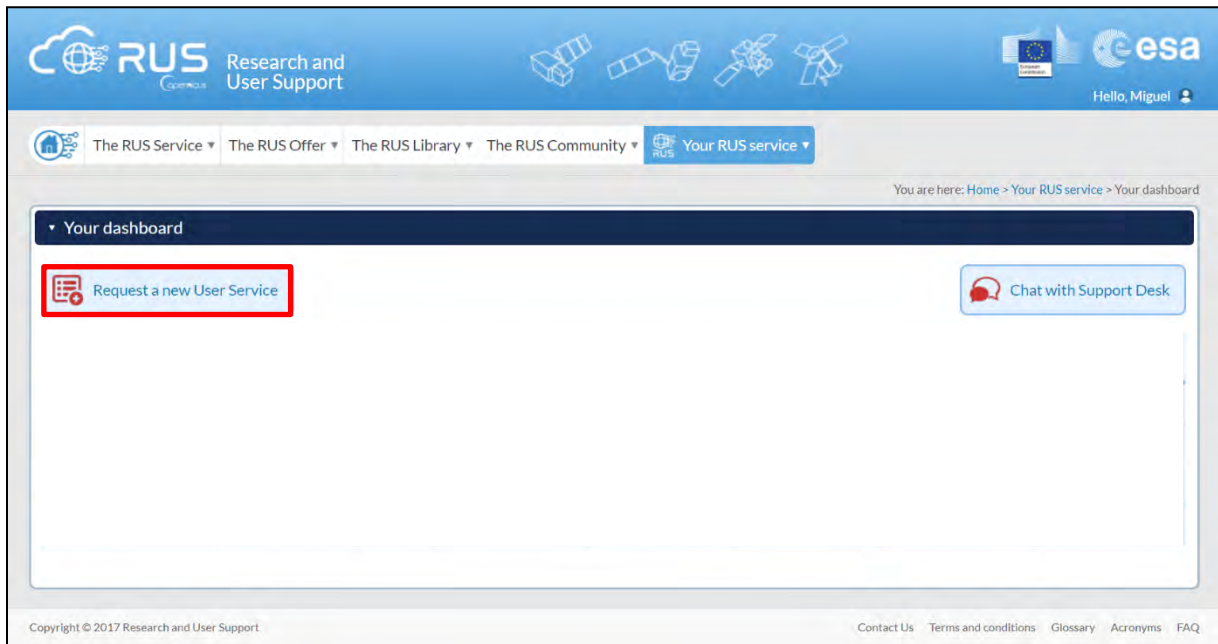
The screenshot shows the RUS dashboard with a modal window titled "Do you want to subscribe for a new RUS account?". The modal contains two sections: "Your ESA-SSO subscription data:" with fields for Login, First Name, Last Name, Email, Organization, and Country; and "Additional subscription information" with a dropdown for "Where did you hear about the RUS service?", a dropdown for "Institution type", a "Phone number" field, and a "Title" field. There are "Subscribe" and "Cancel" buttons at the bottom of the modal.

5 Request a RUS Copernicus Virtual Machine

Once you are registered as a RUS user, you can request a RUS Virtual Machine to repeat this exercise or work on your own projects using Copernicus data. For that, log in and click on **Your RUS Service** → **Your Dashboard**.

The screenshot shows the RUS dashboard with the "Your RUS service" menu open. The menu options are "Your profile", "Your dashboard" (highlighted with a red box), and "Your training". A red arrow points to the "Your RUS service" menu item in the top navigation bar. The dashboard content includes a section for "Your RUS service" with links to "Your profile", "Your dashboard", and "Your training", and a "News from RUS" section with various events and sessions.

Click on **Request a new User Service** to request your RUS Virtual Machine. Complete the form so that the appropriate cloud environment can be assigned according to your needs.



If you want to repeat this tutorial (or any previous one) select the one(s) of your interest in the appropriate field.

The image displays the 'User Support Request' form, specifically Step 1/3 titled 'Your experience'. The form instructs users to provide background information for their User Profile. It includes several questions: 'How many years of experience in Remote Sensing do you have?' with a dropdown menu; 'Have you already downloaded Copernicus data via the Copernicus Open access hubs?' with radio buttons for 'Yes' and 'No'; and 'Have you already handled/processed Copernicus data?' also with radio buttons. A red rectangular box highlights a section asking if the user wishes to practice a tutorial exercise from a RUS webinar, with instructions to hold down the CTRL key for multiple selections. Below this, a list of exercises is provided: HAZA01 - Flood Mapping in Malawi, HAZA02 - Burned Area Mapping in Portugal, HYDR01 - Water Bodies Mapping over Northern Poland, LAND01 - Crop Mapping in Seville, LAND04 - Land Monitoring in Cyprus, and OCEA01 - Ship Detection in Gulf of Trieste. At the bottom, there's a text input field for requesting exercises not in the list, and 'Cancel' and 'Next' buttons.

Complete the remaining steps, check the terms and conditions of the RUS Service and submit your request once you are finished.

User Support Request

Summary information on your request:

This is a collection of information selected across the USR forms.
You can go back and edit this information if necessary.

General information on your request:

Years of experience in Remote Sensing

5-10 years

Downloaded Copernicus data?

✓

Handled/processed Copernicus data?

✓

Webinar codes

HAZA02, LAND04

About your RUS project:

Thematic area

Cryosphere (ice and snow)

Operations to perform on RUS

Algorithm development

Preference for downloading process

Self-downloading

Foreseen activities and support needs

Develop a land cover classification

Project name

RUS_Project1

Earth Observation Data information:

Type of Earth Observation Data:

Sentinel-1

✓

S1 - Product type

S1 - Product 1

S1 - Sensor mode

GRD

S1 - Polarisation

-

S1 - Orbit direction

-

Sentinel-2

X

Sentinel-3

X

Other

X

I don't know

X

Region of Interest:

Min Latitude

39.3303

Max Latitude

40.5877

Min Longitude

-4.6736

Max Longitude

-2.7205

Reference polygons

Data acquisition date(s):

None

Additional data specifications

☒ I have read and agree to the Terms and conditions of RUS Service.

Back and edit

Submit the request

Further to the acceptance of your request by the RUS Helpdesk, you will receive a notification email with all the details about your Virtual Machine. To access it, go to **Your RUS Service** → **Your Dashboard** and click on **Access my Virtual Machine**.

Research and User Support

Hello, Miguel

The RUS Service

The RUS Offer

The RUS Library

The RUS Community

Your RUS service

You are here: Home > Your RUS service > Your dashboard

Your dashboard

Request a new User Service

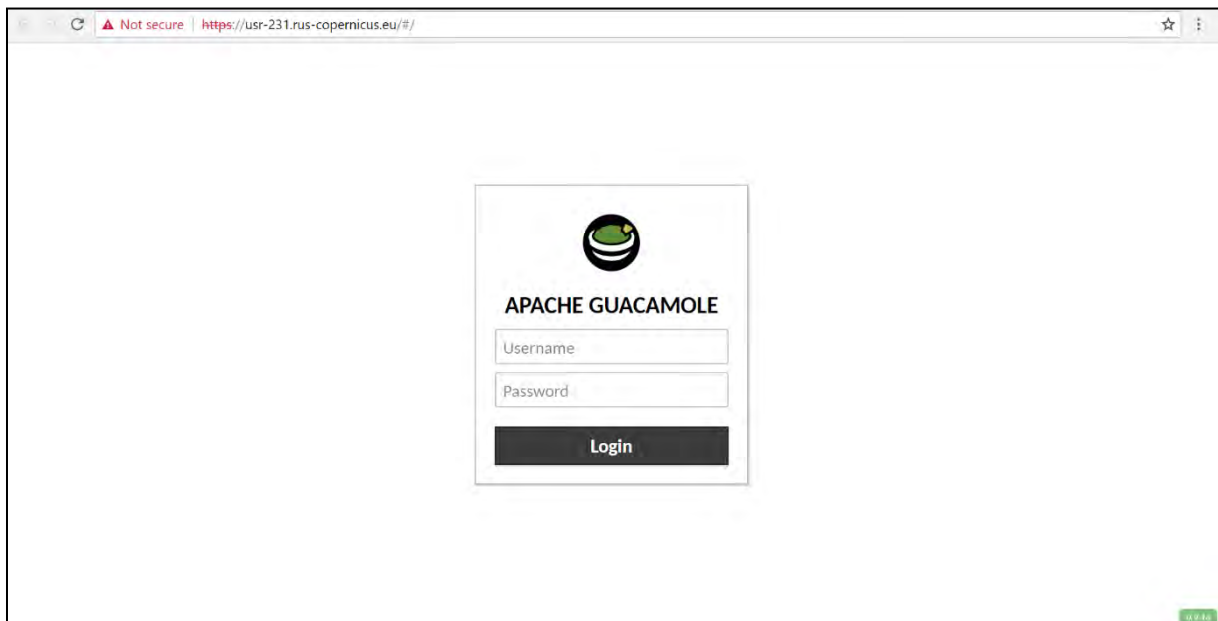
Chat with Support Desk

Project Name	ID	Date of submission	Status	Actions			Virtual Environment	
RUS_training1	231	2017-08-31	Open	Follow my project	Get support	Close my service	Access my Virtual Machine(s)	Access my CPU monitoring dashboard
				Cancel my request	Get a webinar kit	Rate my service ★★★★★	Freeze my Virtual Machine(s)	Report a technical incident

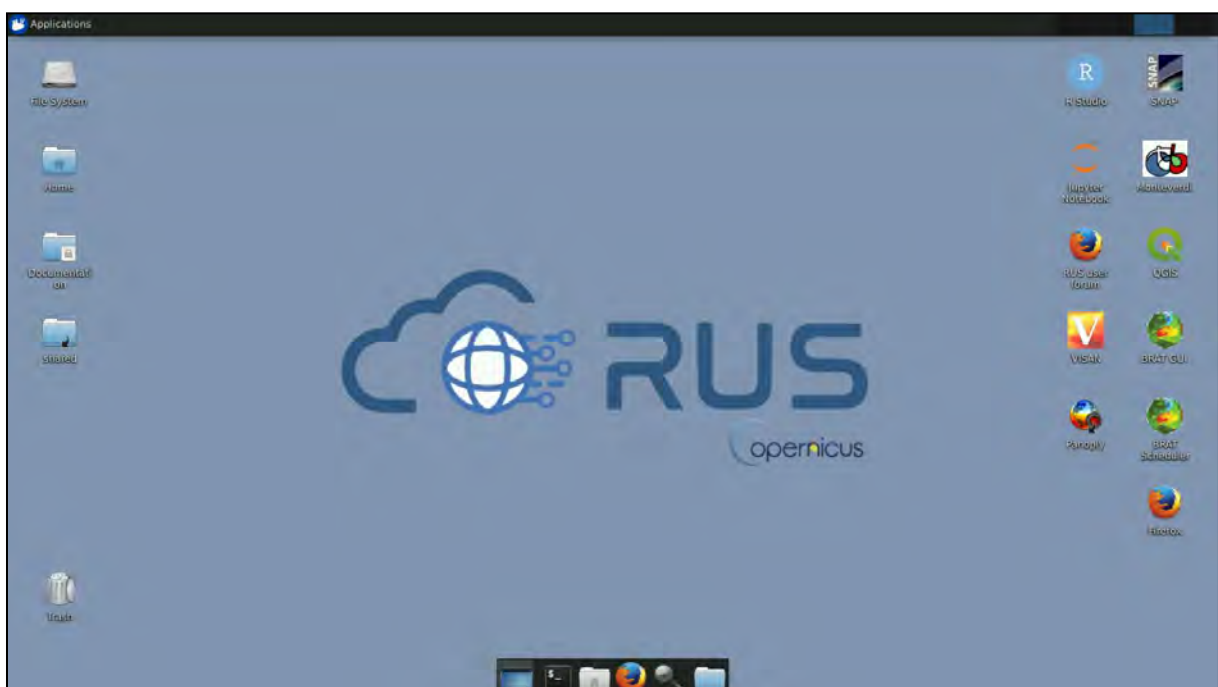
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Contact Us
Terms and conditions
Glossary
Acronyms
FAQ

Fill in the login credentials that have been provided to you by the RUS Helpdesk via email to access your RUS Copernicus Virtual Machine.



This is the remote desktop of your Virtual Machine.

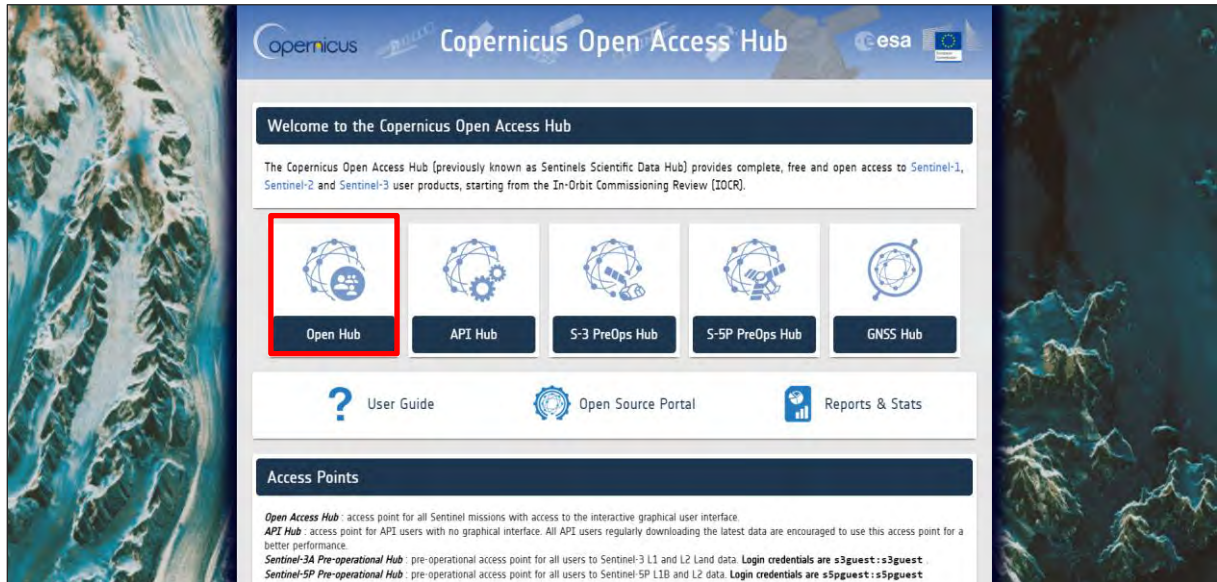


6 Step by step

6.1 Data download – ESA SciHUB

Before starting the exercise, make sure you are registered in the Copernicus Open Access Hub so that you can access the free data provided by the Sentinel satellites.

Go to <https://scihub.copernicus.eu/>



Go to *Open Hub*. If you do not have an account, sign up in the upper right corner, fill in the details and click register.

The image shows the 'Register new account' form on the Copernicus Open Access Hub. The form includes fields for 'Firstname', 'Lastname', 'Username', 'Password', 'Confirm Password', 'E-mail', 'Confirm E-mail', 'Select Domain', 'Select Usage', and 'Select Country'. A red arrow points to the 'REGISTER' button at the bottom right of the form. Another red arrow points to the 'REGISTER' button in the top right corner of the page, which is part of the navigation bar.

You will receive a confirmation email on the e-mail address you have specified: open the email and click on the link to finalize the registration.

Once your account is activated – or if you already have an account – log in.



6.2 Anaconda environment installation

In this exercise we will use *snappy* and python code in JupyterLab to process the Sentinel-1 product we have previously downloaded. However, before starting the analysis, we need to set up both the Anaconda environment and the *snappy* module to work (see  NOTE 1 and  NOTE 2).

 NOTE 1: Project Jupyter is a non-profit, open-source project, born out of the IPython Project in 2014 as it evolved to support interactive data science and scientific computing across all programming languages. Notebook documents (or “notebooks”, all lower case) are documents produced by the Jupyter Notebook App, which contain both computer code (e.g. python) and rich text elements (paragraph, equations, figures, links, etc...). Notebook documents are both human-readable documents containing the analysis description and the results (figures, tables, etc..) as well as executable documents which can be run to perform data analysis. More info at: www.jupyter.org

 NOTE 2: Python is an interpreted, high-level, general-purpose programming language. Created by Guido van Rossum and first released in 1991, Python's design philosophy emphasizes code readability through use of significant whitespace. Its language constructs and object-oriented approach aim to help programmers write clear, logical code for small and large-scale projects. Python is dynamically typed and garbage-collected. It supports multiple programming paradigms, including procedural, object-oriented, and functional programming. Python is often described as a "batteries included" language due to its comprehensive standard library. More info at: www.python.org

Anaconda is a free and open-source distribution of the Python and R programming languages for scientific computing (data science, machine learning applications, large-scale data processing, predictive analytics, etc.), that aims to simplify package management and deployment. Package versions are managed by the package management system *conda*. More info at: <https://www.anaconda.com/distribution/>

To create the same conda environment as the one used for the creation of this training material open Terminal in your RUS Virtual Machine and copy-paste the following command. Then, press *Enter* to run it (this step may take several minutes, do not interrupt it). The conda environment will be called *snappyenv*.

```
conda env create -f /shared/Training/PY01_Sentinel1Processing_snap
py/AuxData/condaenv_PY01.yml
```

Next, we will generate the *snappy* module in that environment so that it can be called and used later.

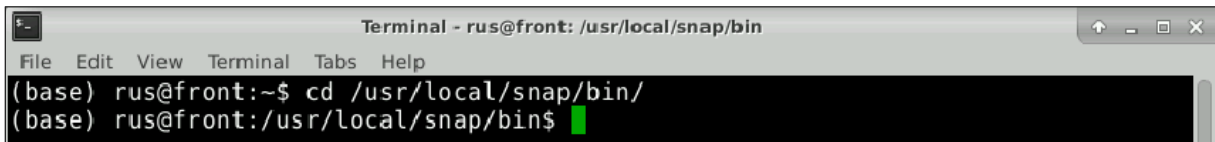
6.3 *snappy* module generation

SNAP provides the Python module *snappy* which allows you to access the SNAP Java API from Python. *snappy* requires either a SNAP installation or a SNAP build. The following instructions will guide you through the installation process to have *snappy* working in the RUS Copernicus Virtual Machine. Generic instructions on how to install *snappy* can be found in the following website:

<https://senbox.atlassian.net/wiki/spaces/SNAP/pages/24051781/Using+SNAP+in+your+Python+programs>

To start, open a Terminal window and navigate to the *bin* folder inside the SNAP installation directory (in RUS Copernicus Virtual Machines - */usr/local/snap/*):

```
cd /usr/local/snap/bin/
```



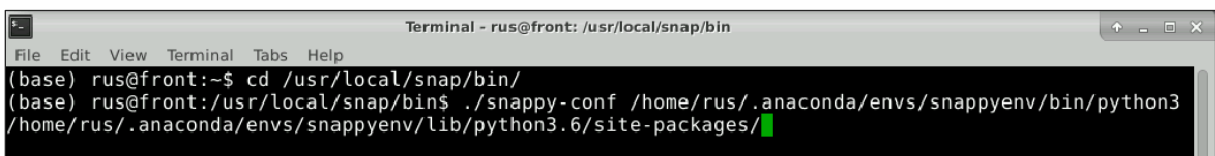
```
Terminal - rus@front: /usr/local/snap/bin
File Edit View Terminal Tabs Help
(base) rus@front:~$ cd /usr/local/snap/bin/
(base) rus@front:/usr/local/snap/bin$
```

Next, we will generate the Python module *snappy* configured for the current SNAP installation and the Python interpreter of choice set in the *<python-exe>* parameter. In addition, instead of generating it in the default folder (*.snap/snap-python* in the home directory), we will place it in the site-package folder of our recently created conda environment. For that, we will make use of the *<snappy-dir>* parameter (see  NOTE 3).

 NOTE 3: The parameter *<python-exe>* must be the full path to the Python interpreter executable which you want to use with SNAP (supported versions are 2.7, 3.3 to 3.6)

Following the previous command in Terminal, copy-paste the next one and press *Enter*.

```
./snappy-conf /home/rus/.anaconda/envs/snappyenv/bin/python3 /home/rus/.anaconda/
envs/snappyenv/lib/python3.6/site-packages/
```



```
Terminal - rus@front: /usr/local/snap/bin
File Edit View Terminal Tabs Help
(base) rus@front:~$ cd /usr/local/snap/bin/
(base) rus@front:/usr/local/snap/bin$ ./snappy-conf /home/rus/.anaconda/envs/snappyenv/bin/python3
/home/rus/.anaconda/envs/snappyenv/lib/python3.6/site-packages/
```

```
Terminal - rus@front: /usr/local/snap/bin
File Edit View Terminal Tabs Help
(base) rus@front:~$ cd /usr/local/snap/bin/
(base) rus@front:/usr/local/snap/bin$ ./snappy-conf /home/rus/.anaconda/envs/snappyenv/bin/python3 /home/rus/.anaconda/envs/snappyenv/lib/python3.6/site-packages/
Configuring SNAP-Python interface...
Done. The SNAP-Python interface is located in '/home/rus/.anaconda/envs/snappyenv/lib/python3.6/site-packages/snappy'
When using SNAP from Python, either do: sys.path.append('/home/rus/.anaconda/envs/snappyenv/lib/python3.6/site-packages')
or copy the 'snappy' module into your Python's 'site-packages' directory.
```

6.4 Sentinel-1 Processing using snappy - JupyterLab

Once the *snappy* module has been properly generated, we can start our exercise. Activate the *snappyenv* conda environment by running the following script in a new Terminal window.

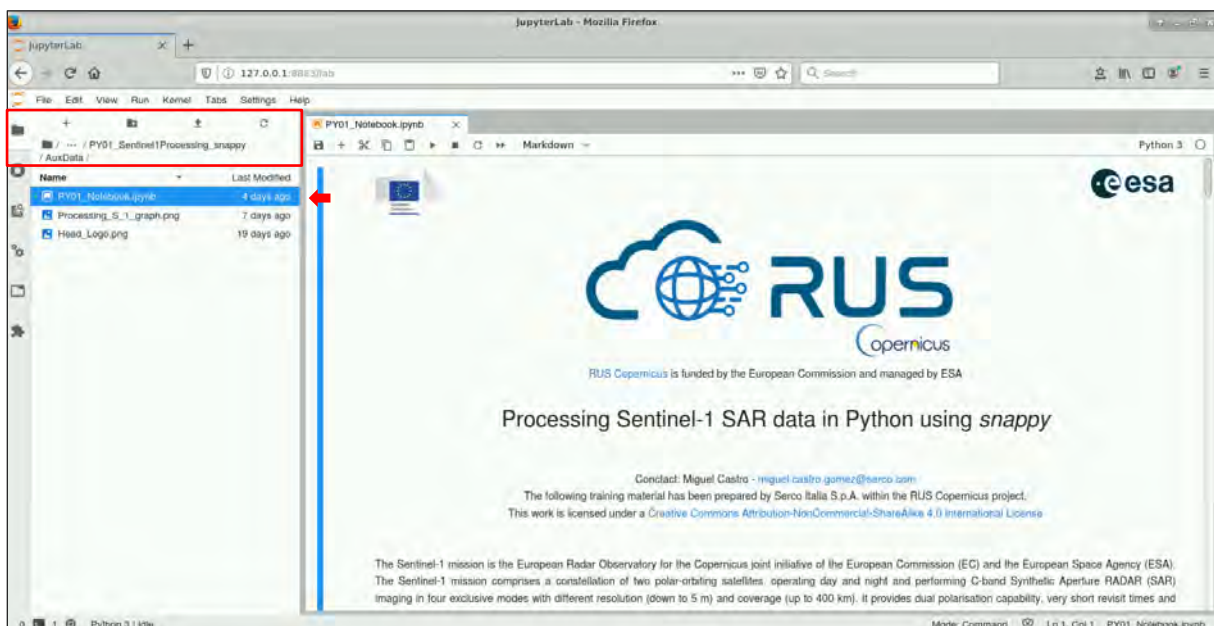
```
conda activate snappyenv
```

Next, write Jupyter Lab and press enter to launch the application. Once open, navigate to the following path inside the JupyterLab GUI and open the file *PY01_Notebook.ipynb*

```
jupyter lab
```

```
Terminal - rus@front: ~
File Edit View Terminal Tabs Help
(base) rus@front:~$ conda activate snappyenv
(snappyenv) rus@front:~$ jupyter lab
```

Path: */shared/Training/PY01_Sentinel1Processing_snappy/AuxData/*



Follow the instructions in the JupyterLab Notebook to continue the exercise.

THANK YOU FOR FOLLOWING THE EXERCISE!

Sentinel-1 User Guide

Sentinel-1 Technical Guide

SNAP GPT Guide

FOLLOW US!!!

